DATA ANALYTICS HUB

Complete Website Documentation & Explanation Guide

Prepared for Shiv Mangal • Educational Data Analytics Website

Project	Data Analytics Hub
Live URL	https://shivmangal66.github.io/data-analytics-hub/
Deployment	GitHub Pages
Frontend	HTML5 + CSS3 + Vanilla JavaScript
Supporting technology	XML sitemap
Browser storage	localStorage
Courses	Excel, Python, SQL, Power BI
Excel path	Day 1–Day 30

1. Project Overview

Data Analytics Hub is a web-based educational platform created to bring Data Analytics learning resources into one place. The homepage introduces the learning paths, and the student dashboard provides course access and progress tracking.

One-line explanation

> “I developed Data Analytics Hub using HTML5, CSS3 and vanilla JavaScript, deployed it on GitHub Pages, and connected learning resources for Excel, Python, SQL and Power BI.”

2. Technologies and Languages Used

HTML5	Page structure: navigation, hero, cards, sections, links and content.
CSS3	Design: colors, gradients, cards, spacing, Grid, buttons and responsive layouts.
Vanilla JavaScript	Dashboard interactions, localStorage, progress calculations and logout.
XML	sitemap.xml for structured URL discovery.
GitHub Pages	Static website hosting/deployment.
localStorage	Browser-side storage for student data and course progress.

3. Website File / Page Structure

- **index.html** — Main landing page with branding, hero section, course overview, features and call-to-action.
- **dashboard.html** — Student dashboard with profile, four courses, PDF links, course tracker and statistics.
- **excel.html** — Excel learning entry page.
- **day1.html to day30.html** — 30 individual Excel learning pages.
- **python-notes.pdf** — Complete Python notes resource.
- **sql-notes.pdf** — Complete SQL notes resource.
- **powerbi-notes.pdf** — Complete Power BI notes resource.
- **sitemap.xml** — Website URL list for search-engine discovery.
- **robots.txt** — Crawler guidance and sitemap reference.

4. Homepage

The homepage uses a dark modern hero design and positions the website as a learning platform for people who want to become Data Analysts. Its source includes Google Search Console verification metadata.

- Sticky navigation with Data Analytics Hub branding.
- Login and Create Account buttons.
- Hero message: Become a Data Analyst from Zero to Pro.
- Learning analytics preview with a visual progress chart.
- Statistics section.
- Four learning paths: Excel, Python, SQL and Power BI.
- Features: Structured Learning, Practical Skills and Track Your Progress.
- Final call-to-action for a free account.

5. Student Dashboard

The dashboard is the main student learning area. It contains a Student Profile, four course cards, a premium-style Course Tracker and overall statistics.

Course cards

- **Excel Data Analytics** — 30 Days Course; Continue Learning opens excel.html.
- **Python Programming** — Complete Notes with Read and Download PDF buttons.
- **SQL** — Complete Notes with Read and Download PDF buttons.
- **Power BI** — Complete Notes with Read and Download PDF buttons.

Student data

The dashboard reads studentName and studentEmail from browser localStorage and places them into the Student Profile when available.

6. Premium Course Tracker

The current dashboard tracks Excel, Python, SQL and Power BI using percentage progress from 0% to 100%. The +10% control increases progress by ten percentage points.

- Progress keys are built as progress_excel, progress_python, progress_sql and progress_powerbi.
- Values are limited to the 0–100 range.
- Progress bars update immediately.
- Status changes between Start learning, Learning in progress and Course completed.
- Overall Progress is the average of the four course percentages.

Technical limitation to explain honestly

“The current tracker is browser-side because it uses localStorage. It does not yet synchronize progress through a central online database.”

7. Course Content

Excel: The notes cover Excel fundamentals, workbook/worksheet structure, data-entry standards, formatting, Tables, navigation and analyst shortcuts. The website also has Day 1–Day 30 pages.

Python: The dashboard connects a complete Python notes PDF with concepts, examples and practical projects. Python is currently presented as complete notes, not as a premium-only course.

SQL: The notes cover databases, SELECT, WHERE, sorting, NULL handling, aggregate functions, GROUP BY/HAVING and advanced topics including joins, window functions, stored procedures, triggers, normalization and query optimization.

Power BI: The notes cover Power BI Desktop, Service and Mobile, data connections, Power Query, data cleaning, data types, merging/appending, data modeling, relationships, DAX, reports and dashboard best practices.

8. How JavaScript Works in the Dashboard

- getItem() reads saved student information from localStorage.
- setProgress() clamps a progress value to 0–100, saves it and updates the UI.
- increaseProgress() adds 10% to the selected course.
- updateTracker() changes the percentage, progress bar and status text.
- updateOverallProgress() calculates the average of the four course percentages.
- logoutUser() removes saved student name/email and redirects to index.html.

9. Responsive Design

CSS media queries adapt the website to smaller screens. The course grid and tracker grid switch from two columns to one column, profile fields stack vertically, and the tracker header changes to a vertical layout.

10. GitHub Pages and Deployment

The project is stored in the GitHub repository Shivmangal66/data-analytics-hub and published through GitHub Pages. HTML pages and static resources such as PDFs, sitemap.xml and robots.txt are part of the project.

Deployment explanation

- Create or update website files in the GitHub repository.
- Commit changes.
- GitHub Pages publishes the static project.
- The published site is available at the GitHub Pages URL.

11. SEO / Google Search Setup

The homepage includes a Google Search Console verification meta tag. The project also includes sitemap.xml and robots.txt. The sitemap lists the homepage, Excel page, Day 1–30 pages and the Python, SQL and Power BI PDF resources.

Important distinction

“A sitemap helps search engines discover URLs, but a sitemap is not a requirement for a website to appear in Google Search.”

12. Current Limitations

- The current project is primarily a frontend/static website.
- There is no server-side database in the current dashboard implementation.
- Student data and course progress are stored in browser localStorage.
- The current PDF links are static resources.
- A secure paid-content system would require a proper payment/access-control implementation; a public PDF URL should not be treated as secure premium delivery.
- The planned ■11 Data Analytics premium product/card is a monetization feature; payment-gateway integration and secure post-payment access are separate implementation work.

13. Ready Answers for Viva / Interview

Q: Which languages did you use?

A: I used HTML5 for structure, CSS3 for styling and responsive design, and vanilla JavaScript for dashboard interactions and progress tracking. XML is used for the sitemap.

Q: Why vanilla JavaScript?

A: The project is lightweight and frontend-focused, so vanilla JavaScript was sufficient for DOM updates, localStorage, progress calculations and logout.

Q: How is progress saved?

A: Course progress is stored in browser localStorage and loaded again when the dashboard opens.

Q: Is there a database?

A: Not in the current version. It uses browser-side localStorage. A future production version could add a backend database for account-based synchronization.

Q: How did you make it responsive?

A: I used CSS Grid and media queries so layouts change automatically for smaller screens.

Q: How did you deploy it?

A: I stored the project in GitHub and deployed the static website through GitHub Pages.

Q: What is the purpose?

A: It provides a single learning platform for core Data Analytics skills—Excel, Python, SQL and Power BI—with a student dashboard and progress tracking.

14. 60-Second Project Presentation

"My project is called Data Analytics Hub. I developed it as an educational website for Data Analytics learners. I used HTML5 for structure, CSS3 for the responsive user interface and vanilla JavaScript for browser-side interactions. The platform provides learning resources for Excel, Python, SQL and Power BI. Excel is organized as a 30-day learning path, while Python, SQL and Power BI are connected as complete notes resources. I also created a student dashboard with profile information and a course tracker that stores progress from 0 to 100 percent in localStorage. The overall progress is calculated from the four course percentages. I deployed the project using GitHub Pages and added Google Search Console verification, sitemap.xml and robots.txt for basic search-engine setup. The current version is frontend/static, so it does not yet use a server-side database or secure payment backend."

15. Quick Revision

HTML5	Structure
CSS3	UI, layout, responsive design
JavaScript	Interactions, localStorage, tracker
XML	Sitemap
GitHub Pages	Hosting / deployment
localStorage	Browser-side data storage
Excel	30-day learning path
Python	Complete notes PDF
SQL	Complete notes PDF
Power BI	Complete notes PDF

End of documentation